

Angular Interview Crash Sheet

Basics

Q: What is Angular?

A: A TypeScript-based front-end framework by Google for building SPAs.

Q: Difference Angular vs AngularJS?

A: AngularJS: JS, MVC. Angular: TS, Component-based, RxJS, AOT.

Q: What are components?

A: Building blocks with template, class, style, metadata.

Q: What are modules?

A: Group of components, directives, pipes, services using @NgModule.

Lifecycle

Q: Component lifecycle order?

A: ngOnChanges → ngOnInit → ngDoCheck → ngAfterContentInit → ngAfterContentChecked → ngAfterViewInit → ngAfterViewChecked → ngOnDestroy.

Q: Constructor vs ngOnInit?

A: Constructor for DI, ngOnInit for init logic.

Q: When use ngOnDestroy?

A: For cleanup: unsubscribing, stopping timers, removing listeners.

Change Detection

Q: What is Change Detection?

A: Process Angular uses to update DOM when data changes.

Q: Default vs OnPush?

A: Default checks all components, OnPush checks only when inputs change.

Q: Pure vs Impure Pipes?

A: Pure runs only when inputs change, Impure runs every cycle.

Routing

Q: What are Route Guards?

A: Services controlling navigation: CanActivate, CanDeactivate, Resolve.

Q: What is Lazy Loading?

A: Load modules on demand using loadChildren.

Advanced

Q: Observables vs Promises?

A: Promises → single value, Observables → multiple values, cancellable.

Q: Subject vs BehaviorSubject?

A: Subject: no initial value, BehaviorSubject: stores latest value.

Q: What are interceptors?

A: Middleware for HTTP requests (auth tokens, logging, errors).

Q: AOT vs JIT?

A: AOT compiles at build-time (faster), JIT compiles at runtime.

Scenarios

Q: Share data between siblings?

A: Use shared service with Subject/BehaviorSubject or parent @Input/@Output.

Q: Unsubscribe observables?

A: Use takeUntil with Subject or async pipe.

Q: Optimize *ngFor?

A: Use trackBy function to avoid re-rendering.

Q: Debounce search input?

A: Use RxJS debounceTime + distinctUntilChanged + switchMap.